

Yue Wu

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EDUCATION



Brown University, Division of Applied Mathematics Providence, RI, USA
Ph.D. Candidate in Applied Mathematics. Advisor: *Prof. Chi-Wang Shu* 08/2023–05/2027 (expected)
M.Sc. in Applied Mathematics. GPA: 4.0 / 4.0 08/2023–05/2025



University of Science and Technology of China, School of the Gifted Young Hefei, Anhui, China
B.Sc. in Information & Computational Science. Rank in major: 1st / 43 09/2019–06/2023

RESEARCH INTERESTS

- High-order numerical methods for partial differential equations
 - Discontinuous Galerkin (DG) finite element and spectral element methods
 - Weighted essentially non-oscillatory (WENO) finite difference and finite volume methods
 - Structure-preserving numerical methods
- Scientific computing
 - Parallel solver development
 - PDE-constrained optimization

PUBLICATIONS AND PREPRINTS

1. **Y. Wu**, S. Becker, J.-M. Qiu, and X. Zhang. Nonnegative low-rank matrix correction under an orthogonality constraint in conservative Vlasov simulations. submitted to *J. Sci. Comput.* arXiv:2607.26272
2. **Y. Wu** and C.-W. Shu. A positivity preserving and entropy stable nodal discontinuous Galerkin scheme for ideal MHD. submitted to *Commun. Math. Stat.* arXiv:2604.23885
3. **Y. Wu** and C.-W. Shu. Finite difference alternative WENO schemes with Riemann invariant-based local characteristic decompositions for compressible Euler equations. *J. Comput. Phys.*, 537:114104, Sept 2025. DOI: 10.1016/j.jcp.2025.114104. MR4912873.
4. **Y. Wu** and Y. Xu. A high-order local discontinuous Galerkin method for the p -Laplace equation. *Beijing J. of Pure and Appl. Math.*, 2(1):373–422, Apr 2025. DOI: 10.4310/BPAM.250415002006.

RESEARCH EXPERIENCE

1. **Structure-Preserving Discontinuous Galerkin Schemes for Magnetohydrodynamics (MHD)**
Brown University. Supervisor: *Prof. Chi-Wang Shu* 03/2025–04/2026
 - Investigated all types of time-explicit structure-preserving conservative schemes for ideal MHD.
 - Developed a nodal discontinuous Galerkin scheme that combines positivity preservation and entropy stability for the first time in literature.
2. **Efficient A-WENO Methods for Compressible Euler Equations**
Brown University. Supervisor: *Prof. Chi-Wang Shu* 09/2024–02/2025
 - Investigated the effect of transform variables in local characteristic decompositions on A-WENO performance and proved the optimal practice.
 - Implemented A-WENO methods using a carefully selected set of Riemann invariants to reduce computational cost.
3. **Discontinuous Galerkin Methods for the p -Laplace Equation**
Bachelor's thesis at USTC. Supervisor: *Prof. Yan Xu* 12/2022–06/2023
 - Proved rigorous a priori error estimates for a high order local DG scheme.
 - Developed an efficient preconditioned gradient descent method.
4. **Positivity-Preserving Conservative Low Rank Methods for Vlasov Dynamics**
Purdue University (remote). Supervisor: *Prof. Xiangxiong Zhang* 06/2022–08/2022
 - Developed multiple nonnegative low rank correction algorithm under orthogonality constraints using convex optimization and tangent space accelerated projection.
 - Incorporate the algorithm into the dynamic low rank Vlasov solver as a positivity preserving conservative limiter to obtain robust and physically meaningful simulation.

5. Numerical Simulation of Plasma Equilibrium Evolution in Nuclear Fusion

USTC undergraduate research project. Supervisor: *Prof. Mengping Zhang*

06/2021–05/2022

- Developed a parallel hybrid finite difference-pseudo spectral code for resistive MHD in toroidal geometry, and performed long-time simulation of resistive tearing mode instability in tokamaks.
- Validated the results with researchers from the Institute of Plasma Physics, CAS, and against those from existing open-source codes.

TEACHING EXPERIENCE

1. TA @ Brown: <i>APMA 0355—Applied ODEs with Theory</i>	Fall 2026
2. TA @ Brown: <i>APMA 1655—Intro. to Prob. & Stats. with Theory</i>	Spring 2026
3. TA @ Brown: <i>APMA 0160—Intro. to Sci. Computing</i>	Fall 2025
4. TA @ Brown: <i>APMA 1650—Stat. Inference I</i>	Spring 2025
5. TA @ Brown: <i>APMA 1210—OR: Deterministic Models</i>	Fall 2024
6. TA @ USTC: <i>Computational Methods B</i>	Spring 2022

PRESENTATIONS

1. Poster session, the 2024 International Congress of Basic Science (ICBS), Beijing, China	07/2024
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PROFESSIONAL SERVICES

1. Jorunal reviewer for <i>J. Comput. Phys.</i> and <i>J. Sci. Comput.</i> (since 2025), and <i>Comput. Math. Appl.</i> (since 2026)
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PROFESSIONAL SKILLS

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- Programming: C++, MATLAB, Python, Fortran 2008, MPI, OpenMP
 - Software: NGSolve, FEniCS, MFEM, Mathematica, Tecplot
 - Tools: Linux, Git, Jupyter, L^AT_EX, Markdown
 - Language: Mandarin Chinese, English

HONORS AND AWARDS

• Howard and Jan Swearer Graduate Fellowship	AY 2023–2024
• USTC Outstanding Undergraduate Award	06/2023
• “Chia-Chiao Lin” Gold Medal (Nation Top 1) in Applied and Computational track & Team Silver Medal & Excellence Prize in Analysis and PDEs track, the 14th S.-T. Yau College Student Mathematics Contest	06/2023
• Gold Prize, USTC Outstanding Student Scholarship	10/2022
• Excellence Prize in Analysis and PDEs track, the 13th S.-T. Yau College Student Mathematics Contest	08/2022
• China National Scholarship	12/2021
• Second Prize, the 13th Chinese Mathematics Competitions	12/2021
• China National Scholarship	12/2020
• Third Prize, USTC Freshman Scholarship	09/2019

EXTRACURRICULAR ACTIVITIES

• USTC road cycling team member, USTC	09/2019–06/2023
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last update: September 22, 2026